

V6 Phase 1 Delivered. Pattern Holds for 4 Rounds.

Adaptive Nudges + Evolution Tracker built and wired. Next: Background Adaptation + Trajectory Intervention.

V6 Phase 1 — Verification

#	Spec	Backend	API	Frontend	Status
1	Adaptive Nudge Engine	193 lines, 3 nudge sources	200, 3 nudges	TODAY nudge card with Accept/Dismiss (8 refs)	DELIVERED
2	Evolution Tracker	195 lines, 4 failure-mode sources	200, 1 failure mode	LEARN "Mistakes no longer made" section	DELIVERED

V6 PHASE 1: DELIVERED. PATTERN HOLDS FOR 4 ROUNDS.

Commit `dd818aa`. 476 insertions across 5 files. 2 new backend modules (388 lines: `adaptive_nudge.py` 193, `evolution_tracker.py` 195). Both APIs return 200 with real data. Both wired to existing frontend surfaces (TODAY for nudges, LEARN for evolution tracker). The "built but not applied" pattern has NOT recurred for 4 consecutive rounds (rounds 16, 18, 21, 23). This is the longest streak in the engagement.

V5 + V6 litmus tests: PASS. Sidebar: 4 items (unchanged). No new panels. No new sidebar items. Nudge card in TODAY (8 refs). Evolution tracker in LEARN (1 ref). Both enhance existing surfaces. V5: UI simpler. V6: permanently improves the organization (nudges restructure work; tracker measures elimination).

Test suite: 423 pass, 0 fail, 2 skipped. (389 API+auth + 34 frontend+cognitive). No regressions.

Nudge quality: 3 nudges generated. Nudge 1 (Globex buying committee): evidence references causal data ("produced the same positive outcome 85 times"). Intervention is actionable. Nudge 2 (priya.m bottleneck): evidence references causal data ("81 times"). BUT intervention is generic ("Proactively address the bottleneck") — not specific enough per the spec. Nudge 3 (deployment concentration): structural evidence (not causal — "only priya.m holds knowledge"). Intervention is specific ("Cross-train carlos.r on deployment"). Mixed quality — 2 of 3 nudges have causal evidence; 2 of 3 have actionable interventions; 1 of 3 has both. Pilot polish: improve the `_derive_intervention()` function to produce specific restructuring for all nudge types.

Evolution tracker quality: 1 failure mode tracked (commitment integrity gap, status: active). 0 eliminated (honest: "the pilot is too young — needs 90+ days without recurrence"). This is correct honest degradation. `frequency_history` has 0 entries (the demo data does not have enough temporal resolution for per-period frequency). Minor data quality issue, not a blocker.

Score: V6 Phase 1 DELIVERED. Both specs built, wired, verified. The product now has 15 cognitive engines across 6 constitutional versions. The organization can now be nudged toward better work patterns and its mistakes can be tracked toward elimination. Next: V6 Phase 2-3.

V6 Phase 2-3 — Strict Coding Instructions (Next Task)

V6 Phase 1 is complete. The next task is V6 Phase 2-3: Background Adaptation + Trajectory Intervention + Organizational DNA + Evolution Narrative. 4 specs, ~10.5 days. Build in order.

V6 Spec #3 — Background Adaptation Loop (Build First, 2 days)

What: V6 Law 2: "The organization should become more intelligent even when nobody opens Maestro." The Background Adaptation Loop runs on every signal ingest (not on user request), checks for improvement opportunities, and queues nudges for the next interaction.

Build:

1. CREATE `backend/maestro_oem/background_adaptation.py` — the BackgroundAdaptationLoop. On every signal ingest (hooked into `oem_state.py live_ingest()`), the loop: (a) checks if the new signal creates a new pattern that warrants a nudge, (b) checks if any active nudge should be escalated (problem worsening), (c) checks if any resolved failure mode is recurring (regression detection), (d) queues any generated nudges for the next user interaction.
2. CREATE GET `/api/oem/adaptation/status` endpoint.
3. MODIFY `backend/maestro_api/oem_state.py live_ingest()` — call the background loop after each signal batch.
4. MODIFY `static/js/today.js` — if background-detected nudges exist, show them with a "Maestro noticed this while you were away" label.

Acceptance test: 1) GET `/api/oem/adaptation/status` returns `background_nudges` + `regression_alerts` + `escalation_alerts`. 2) The loop runs on signal ingest (check `oem_state.py` calls it). 3) TODAY shows "while you were away" label for background nudges. 4) V5 litmus: no new panel. 5) V6 litmus: runs in background (not on user request).

V6 Spec #4 — Trajectory Intervention (Build Second, 2.5 days)

What: Weak signal -> trajectory change -> quiet intervention -> failure prevented. The user never realizes something was prevented. "Trust between Engineering and Legal is declining. If unchecked, coordination will fail within 3 weeks. A joint review session reversed it before."

Build:

1. CREATE `backend/maestro_oem/trajectory_intervention.py` — composes `trajectories.py` (V5 #7, exists) + `institutional_recall.py` (V5 #8, exists) + `adaptive_nudge.py` (V6 #1, exists). Detects: trajectory declining, projected threshold crossing, `time_to_failure`, historical analogue, intervention.
2. CREATE GET `/api/oem/interventions` endpoint.
3. MODIFY `static/js/today.js` — if a trajectory intervention exists with high urgency, show it as the TOP item: "Maestro detected a risk..."

Acceptance test: 1) GET `/api/oem/interventions` returns `interventions` array (or honest "stable"). 2) Each has trajectory, `time_to_failure`, intervention, `evidence_count`, urgency. 3) `time_to_failure` computed from trajectory slope. 4) TODAY shows trajectory intervention as top item when urgency is high. 5) V5 litmus: no new panel. 6) V6 litmus: prevents failures before they occur.

V6 Spec #5 — Organizational DNA (Build Third, 3 days)

What: "This is what YOUR organization would do when it is at its best." Not industry best practice — the org's own best self. 7 chromosomes that evolve over time and drive recommendation filtering.

Build:

1. CREATE `backend/maestro_oem/organizational_dna.py` — extends `personality.py` (V3, exists) with: evolution tracking (then vs now), decision-style inference, recommendation alignment filtering.

2. 7 chromosomes: decision_style, risk_appetite, learning_velocity, communication_style, conflict_style, innovation_style, execution_style. Each with value, confidence, evidence_count, basis.

3. CREATE GET /api/oem/dna endpoint.

4. MODIFY backend/maestro_oem/wisdom.py — filter recommendations by DNA alignment ("This aligns with your consensus-driven style").

5. MODIFY static/js/learn.js — add "Who your organization has become" section showing DNA evolution.

Acceptance test: 1) GET /api/oem/dna returns 7 chromosomes with value + confidence + evidence_count. 2) At least 1 evolution entry (then/now/narrative). 3) recommendation_alignment includes a real rec_id. 4) wisdom.py references DNA alignment. 5) LEARN shows DNA section. 6) V5 litmus: no new panel. 7) V6 litmus: DNA evolves and drives better-aligned recommendations.

V6 Spec #6 — Evolution Narrative Engine (Build Last, 2 days)

What: The organization's autobiography. Not CRM, not Slack history — its autobiography. Chapters with titles, periods, narratives, lessons. "Your organization started fast but fragile. It learned that review produces better outcomes."

Build:

1. CREATE backend/maestro_oem/evolution_narrative.py — composes organizational_dna.py (V6 #5, just built) + evolution_tracker.py (V6 #2, exists) + identity.py (V4, exists) + skepticism.py (V4, exists) + principles.py (V4, exists).

2. Produces chapters: {title, period, narrative (3+ sentences), lessons (2+), evidence_count}. Plus overall_story and next_chapter_prediction.

3. CREATE GET /api/oem/autobiography endpoint.

4. CREATE static/js/autobiography.js — surface (command-palette only, NOT in sidebar). Renders the autobiography as a calm narrative.

Acceptance test: 1) GET /api/oem/autobiography returns chapters (2+ or honest "first chapter"). 2) Each chapter has title, period, narrative (3+ sentences), lessons (2+), evidence_count. 3) overall_story and next_chapter_prediction non-empty. 4) Autobiography surface accessible via Ctrl+K. 5) V5 litmus: command-palette only (no new sidebar item). 6) V6 litmus: makes the institution's evolution visible and irreversible.

Build Order

Step	Spec	Effort	Frontend Surface	Key Dependency
1	#3 Background Adaptation	2 days	TODAY (while-you-were-away label)	oem_state.py live_ingest() hook + adaptive_nudge.py (V6 #1)
2	#4 Trajectory Intervention	2.5 days	TODAY (top item when urgent)	trajectories.py (V5 #7) + recall.py (V5 #8) + adaptive_nudge.py (V6 #1)
3	#5 Organizational DNA	3 days	LEARN (who you've become)	personality.py (V3) + wisdom.py (V4)
4	#6 Evolution Narrative	2 days	Autobiography surface (Ctrl+K only)	organizational_dna.py (V6 #5) + evolution_tracker.py (V6 #2) + identity.py (V4)
TOTAL	4 specs	~10 days	TODAY + LEARN + Autobiography (all existing or V6 Phase 1 complete)	

Strict Rules for V6 Phase 2-3

Rule	Enforcement
1. Build in order: #3 -> #4 -> #5 -> #6	Dependencies are real. #4 needs #3 (background nudges). #6 needs #5 (DNA) + #2 (tracker).
2. Background Adaptation MUST hook into live Nudge-triggered API. Must run on signal ingest. V6 Law 2: "even when nobody opens Maestro."	
3. Trajectory Intervention MUST compute time. Not included in the trajectory slope (from trajectories.py) projects when the dimension crosses the danger threshold.	
4. DNA MUST filter recommendations (wisdom.py) just display. wisdom.py must reference DNA alignment when synthesizing judgment. "This aligns with your consent."	
5. Autobiography is command-palette ONLY	NOT in the sidebar (stays at 4 items). V5 litmus: no new sidebar items.
6. V5 litmus: UI SIMPLER	No new panels in sidebar. New content enhances TODAY, LEARN, or command-palette surfaces.
7. V6 litmus: permanently improves the organization. #3: background adaptation (runs without user). #4: prevents failures. #5: DNA evolves and drives recommendations. #6: improves the organization.	
8. 5-point checklist (all YES)	1) Full acceptance test. 2) Application not existence. 3) Full test suite. 4) Live API. 5) UI simpler + org permanently improved.
9. No silent skips	If a spec is deferred, say so explicitly. Do NOT claim "all 4 delivered" if only 3 are built.
10. Run FULL test suite	423+ tests. Report exact count. No subsets.

Final Charge

V6 PHASE 1 DONE. PHASE 2-3 NEXT.

V6 Phase 1 is delivered. Adaptive Nudges restructure work. Evolution Tracker measures eliminated mistakes. The pattern has held for 4 consecutive rounds. The test suite is green. 15 cognitive engines across 6 constitutional versions, all invisible, all enhancing existing surfaces.

Build V6 Phase 2-3 next. 4 specs, ~10 days. Background Adaptation makes Maestro active when nobody is looking. Trajectory Intervention prevents failures before they occur. Organizational DNA encodes the institution's evolving judgment. Evolution Narrative produces the autobiography. Build in order: #3 -> #4 -> #5 -> #6.

The V6 constitutional law: "Every interaction must permanently improve the organization." Background adaptation improves without interaction. Trajectory intervention prevents failures. DNA evolves and filters recommendations. Autobiography makes evolution irreversible. All four permanently improve the organization. Build them.

After V6 Phase 2-3: V6 is complete. Then V7.1 (self-revising institutional model with Bayesian competing hypotheses). The path is clear. Build V6 Phase 2-3. Then V7.1. Then ship the 90-day pilot. The Invisible Layer + Institutional Adaptation + Self-Revising Model = an operating system for institutional learning. Build it.